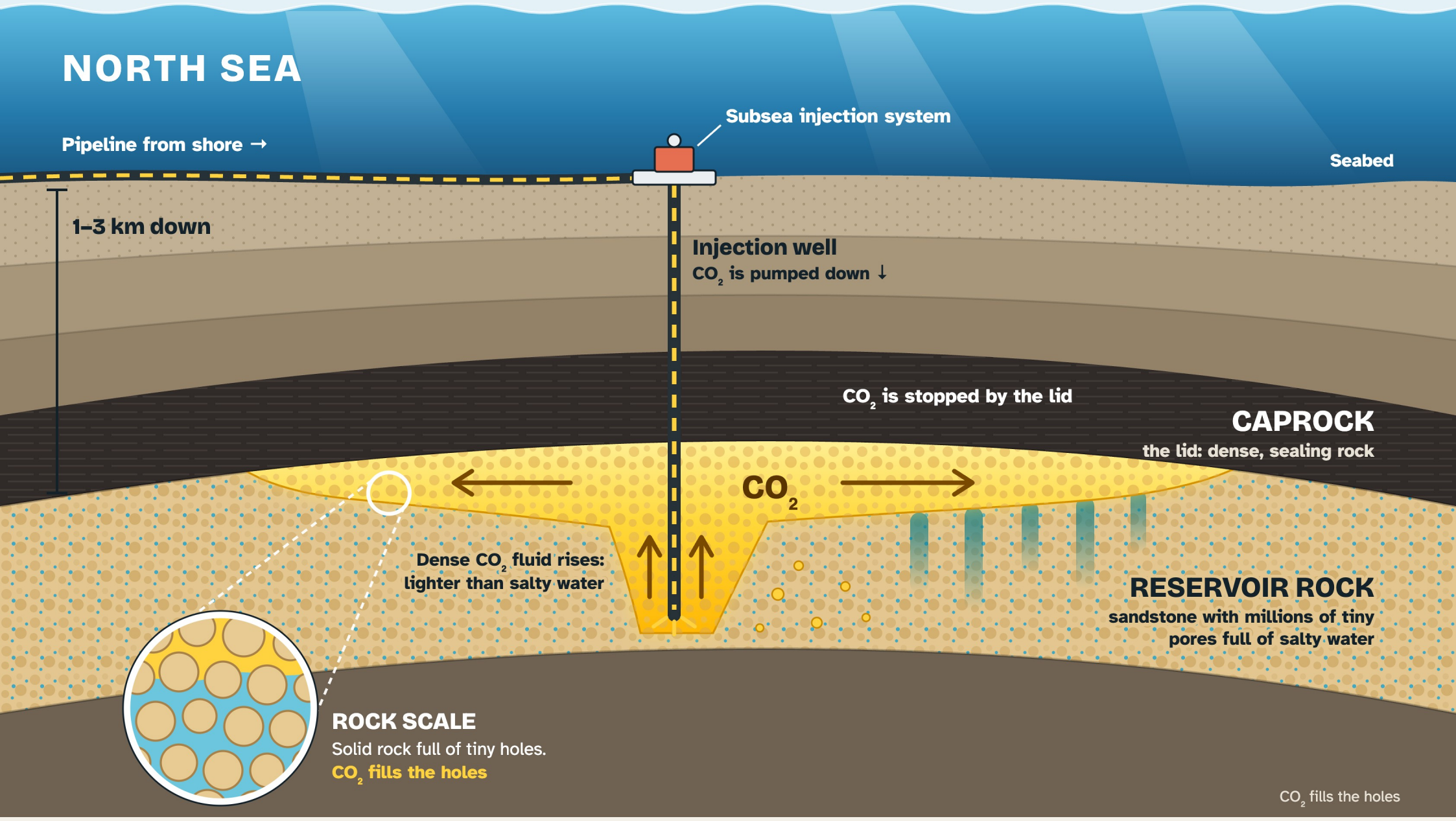
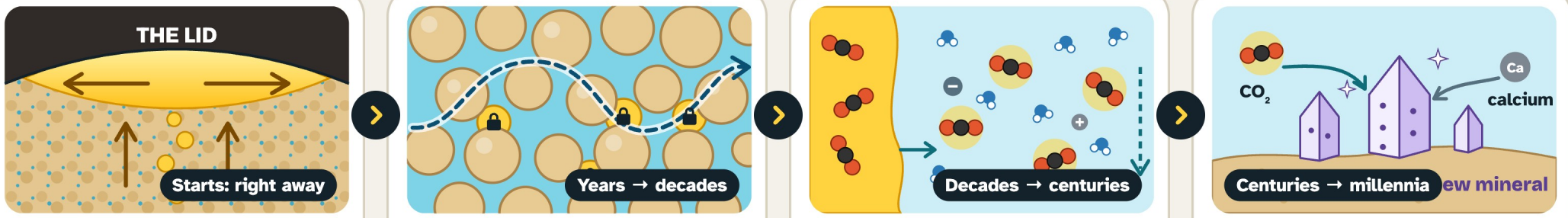
CARBON CAPTURE & STORAGE · NORTH SEA

WHERE DOES THE CO_2 GO?

Discover how CO_2 can stay trapped deep beneath the North Sea.



THE 4 WAYS CO_2 GETS TRAPPED



1 STRUCTURAL TRAPPING

A rock layer acts like a lid.

CO_2 rises because it is lighter than water, then collects under an impermeable sealing rock.

ZOOM 1 · RESERVOIR

2 RESIDUAL TRAPPING

Tiny CO_2 bubbles get stuck.

As water flows back in, small blobs of CO_2 are held between rock grains by capillary forces.

ZOOM 2 · PORES (<1 mm)

3 SOLUBILITY TRAPPING

CO_2 dissolves into underground water.

CO_2 molecules mix into the salty water. That water gets slightly heavier and slowly sinks.

ZOOM 3 · MOLECULES

4 MINERAL TRAPPING

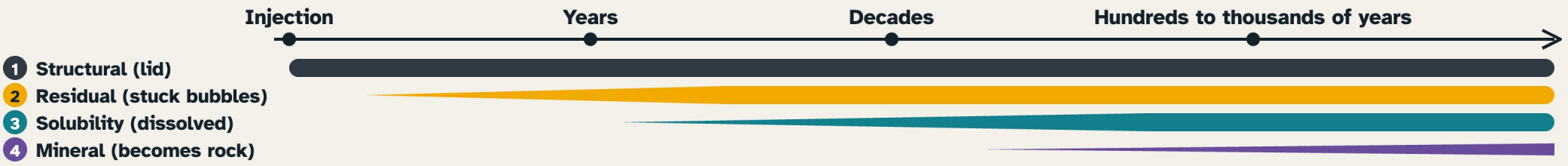
CO_2 can eventually become part of the rock.

Dissolved CO_2 slowly reacts with the rock to form solid carbonate minerals, such as calcite.

ZOOM 4 · MINERALS

CO_2 becomes increasingly trapped through multiple natural processes underground.

MORE SECURE →



One reservoir. Four trapping mechanisms.
Increasing security over time.

Caprock → Trapped bubbles → Dissolved CO_2 → Minerals